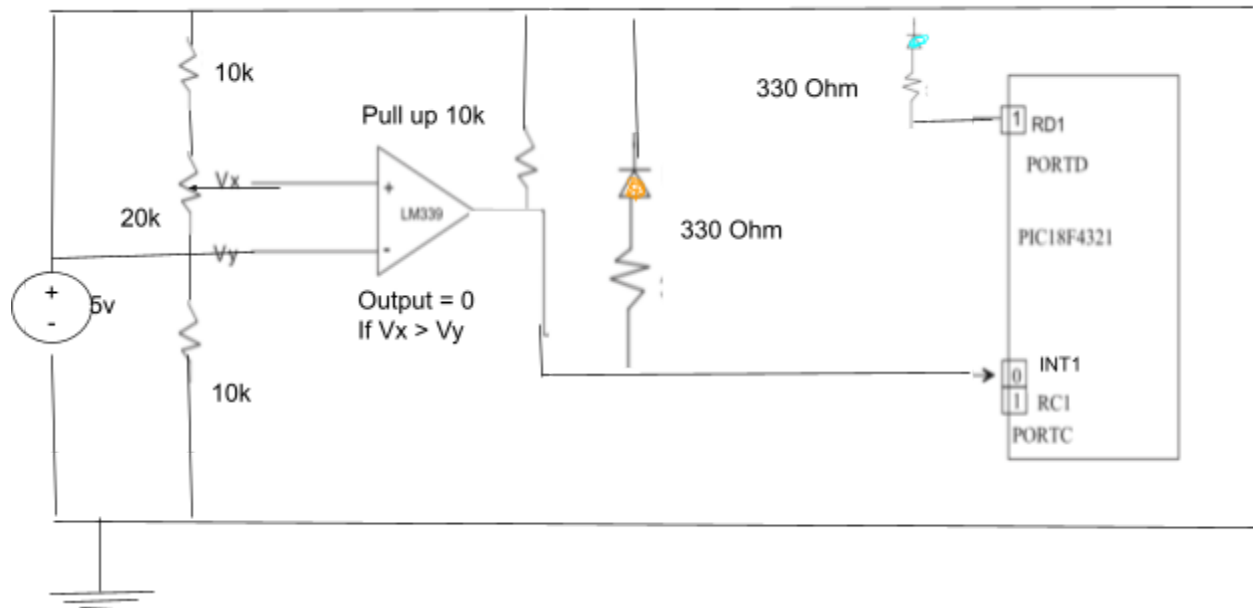
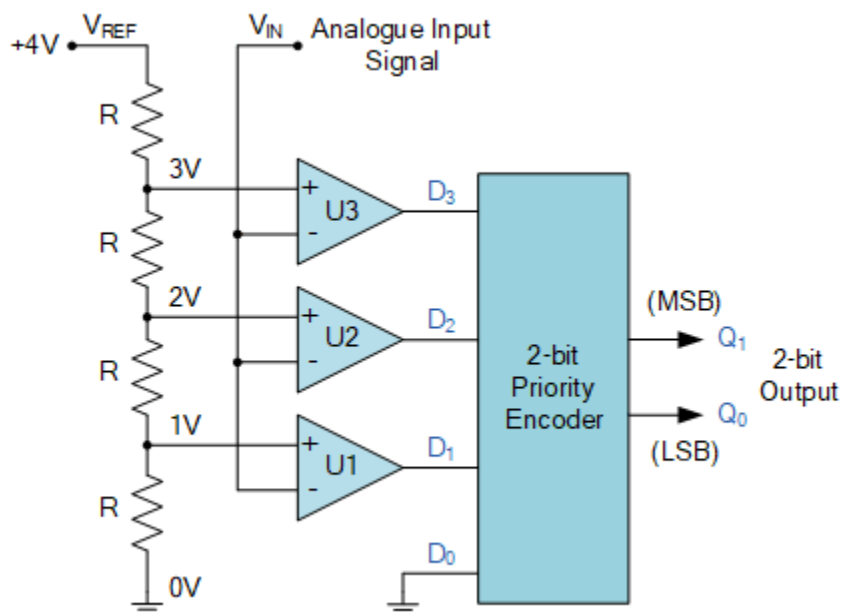


Sarah Boyd
Group members: Wesley Hwee and Mariana Rivera
ETE 4201
EXTRA CREDIT

ADC Comparator Circuit EXTRA CREDIT

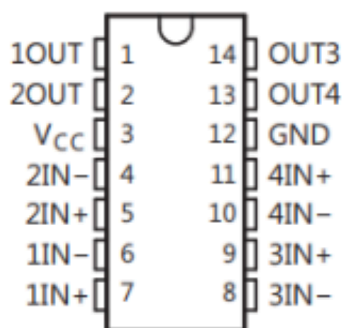


IMG 1. The schematic of the circuit

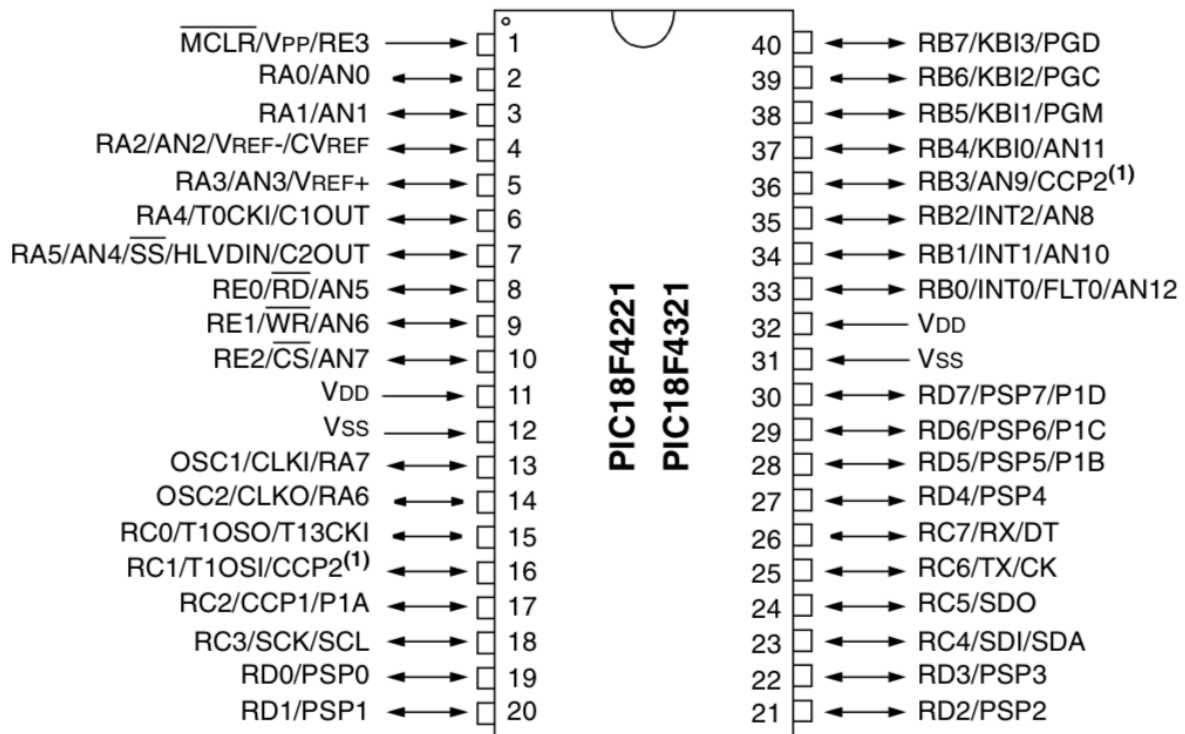


IMG 2. An example of a comparator, found at electronics-tutorials.ws/combination/analogue-to-digital-converter.html

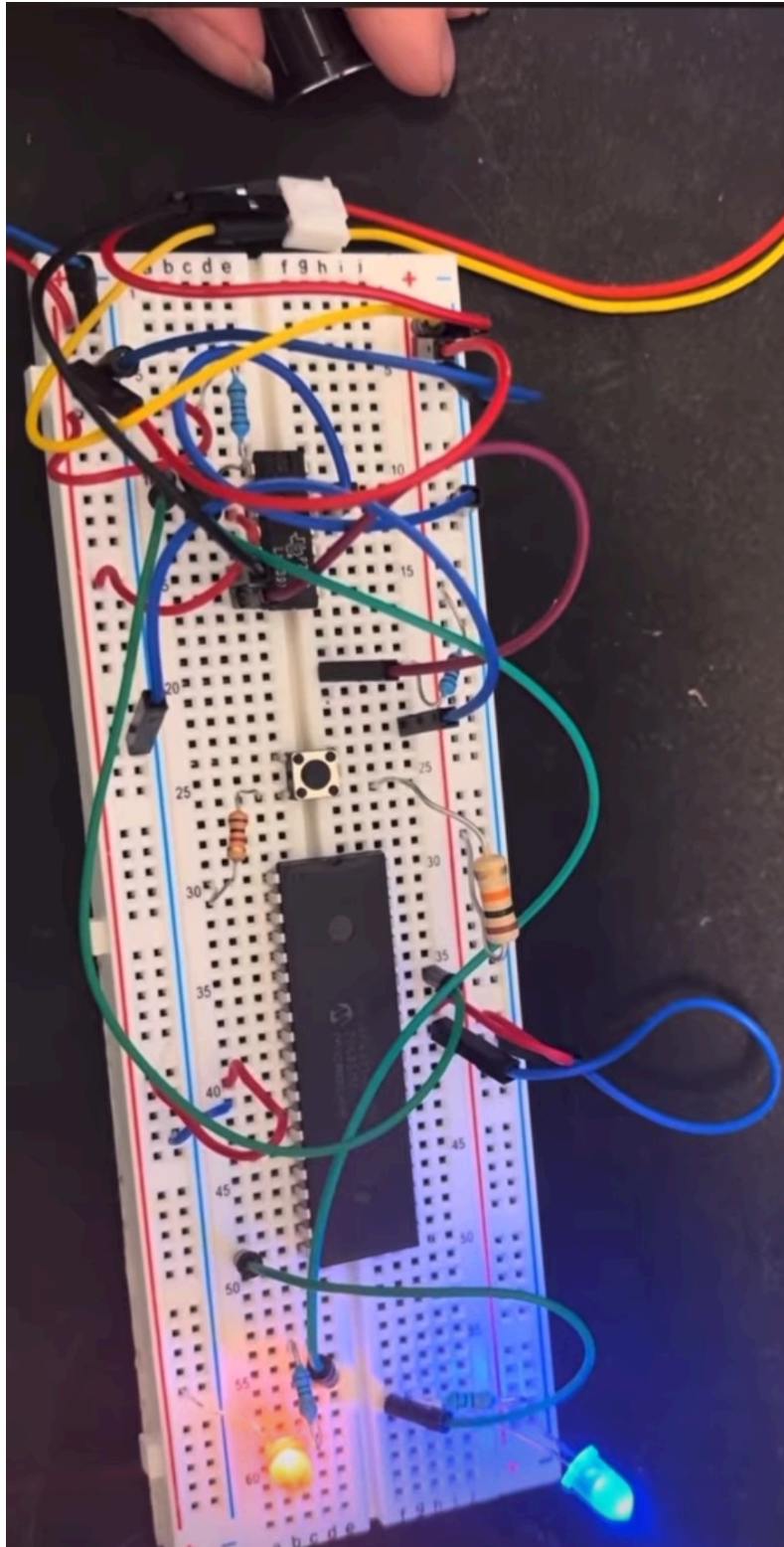
The 5v serves as v_y , and is also the power for pins 11 and 32 of the PIC18F, and pin 3 of the LM399n. 5 volts is essentially the full scale reference voltage (V_{ref}) mentioned in class, and v_y would be modified with resistors to be a specific value for comparison. In an analog to digital converter, there are several voltages made from the reference voltage (amounts determined by a voltage divider, and controlled with resistors) and could be amounts like 3v, 2v, 1v, 0v, etc. V_x , the variable voltage controlled by a $20k\Omega$ potentiometer, represents the real life voltage of a circuit. This voltage is to be measured and converted into a digital format to be observed. When $V_x > V_y$, the comparator outputs a 0. Otherwise, the comparator outputs 5 volts (logic 1). In a real analogue to digital converter, there is a series of comparators comparing our measured voltage (V_x) to various V_y voltages. In my circuit, when $v_x > v_y$ the comparator outputs a 0, turning on the orange LED and sending that same output of 0 to the Pic18F. The blue LED is the output of the Pic18F- if bit 1 of port D is 0, then the LED will turn on. In the case of a converter, an output of 0 or 1 from a specific converter indicate a logic 0 or 1 of the voltage of that amount being there. With enough steps (bits) the digital output can approximate the analog voltage more finely, basically creating a digital wave form of “stairs”. Note that a voltage of 4.99 is still rounded down to 4v. With enough of these “stair steps”, we can make an approximate waveform representing the measured voltage.



IMG 3. Layout of the LM339n comparator, from Ovega.com



IMG 4. Layout of the PIC18F4321, from microchip.com.



IMG 5. *This circuit was demonstrated in class and approved, but additionally the video explanations and demonstrations of this circuit are provided below.*

Explanation of circuit: https://youtu.be/xm27ZBMHhvQ?si=nxL_SgcG9XJujM9W

Demonstration of
circuit: <https://youtube.com/shorts/b88cyDDN-xU?si=5HeaF3Qd2cH8mU8D>